Fiscal Year:	FY 2015	**Task Last Updated:**	FY 01/13/2015
PI Name:	Volz, Martin Ph.D.		
Project Title:	Reduction of Defects in Germanium-Silicon (RDGS)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	256-544-5078
Organization Name:	NASA Marshall Space Flight Center		
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City:	Huntsville	**State:**	AL
Zip Code:	35812	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	03/06/2008	**End Date:**	10/15/2018
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Downey, James1 Patton	**Contact Phone:**	
Contact Email:	patton.downey@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: end date change per P. Downey/MSFC (Ed., 1/12/2015) NOTE: end date change per P. Downey/MSFC (Ed., 12/21/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mazuruk, Konstantin (University of Alabama Huntsville) Croell, Arne (University of Freiburg)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Original Principal Investigator was Dr. Frank Szofran, Feb 1998-Jan 2004; project continues with Dr. Martin Volz. It is well established that crystals grown without contact with a container have far superior quality to otherwise similar crystals grown in direct contact with a container. In addition to float-zone processing, detached-Bridgman growth is often cited as a promising tool to improve crystal quality, without the limitations of float zoning. In detached growth, most of the melt has contact with the crucible wall, but the crystal is separated from the wall by a small gap, typically 10-100 microns. A liquid meniscus extends between the top edge of the growing crystal and the crucible wall. Detached growth has been found to occur quite often during microgravity experiments and considerable improvements of crystal quality have been reported for those cases. However, no thorough understanding of the process or quantitative assessment of the quality improvements existed prior to the investigation. This project will determine the means to reproducibly grow Ge-Si alloys in the detached mode. Ge-Si alloys are of technological interest because with lower defect density materials it would be possible to make thermal neutron and x-ray optics, faster devices than with silicon, higher efficiency solar cells, and higher thermopower devices. Although these alloys are model materials in the sense that these results should be applicable to many other materials, the major technological significance is that the same production equipment that is currently used for silicon device manufacture can also be used for germanium-silicon devices. This means that using germanium-silicon instead of some other material to achieve improved device performance will save an immense amount of money in development and purchase of production equipment. Several tasks have been completed or are planned as part of the investigation: * measure the relevant material parameters such as contact angle, growth angle, surface tension, and wetting behavior of the GeSi-melt on potential crucible materials; * determine the mechanism of detached growth including the role of convection; * quantitatively determine the differences in defects and impurities among normal Bridgman, detached Bridgman, and float zone (FZ) growth; * investigate the influence of defined azimuthal or meridional flow due to rotating magnetic fields on the characteristics of detached growth; * control time-dependent Marangoni convection in the case of FZ-growth by the use of a rotating magnetic field to examine the influence on the curvature of the solid-liquid interface and the heat and mass transport; and * grow high quality GeSi single crystals with Si concentrations up to 10%. Determine the necessary conditions for reproducibly achieving detachment of Ge-Si alloys in the Bridgman growth method, and quantitatively compare the defect structures and impurity levels of detached Bridgman, normal Bridgman, and floating-zone grown materials.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will increase our understanding of the mechanisms which lead to crystal detachment and enhance our ability to control detachment both in microgravity and on Earth. Control of detachment will enable the growth of relatively large diameter crystals with orders of magnitude fewer defects than can be achieved with standard growth techniques. The improvement in material quality may lead to better performance for a variety of electronic devices.
Task Progress:	A study was completed analyzing the shape evolution of detached Bridgman crystals grown in microgravity. Detached Bridgman crystal growth defines a process in which a gap exists between the growing crystal and the crucible wall. In microgravity, the parameters that influence the existence of a stable gap are the growth angle of the solidifying crystal, the contact angle between the melt and the crucible wall, and the pressure difference across the meniscus. During actual crystal growth, the initial crystal radius will not have the precise value required for stable detached growth. Beginning with a crystal diameter that differs from stable conditions, numerical calculations were used to analyze the transient crystal growth process. Depending on the initial conditions and growth parameters, the crystal shape either evolves towards attachment at the crucible wall, towards a stable gap width, or inwards towards eventual collapse of the meniscus. Dynamic growth stability was observed only when the sum of the growth and contact angles exceeds 180°. The study was published in Crystal Research and Technology 49, 768-776 (2014). Ground-based testing of ampoule designs continued. The different designs are required to accommodate the variation in processing parameters in the flight experiment matrix. These parameters include ampoule material, sample material, and internal ampoule pressure. In some experiments the sample stuck to the SiO2 ampoule material and alumina (Al2O3) is being considered as a replacement. Characterization techniques are being developed for the ground test samples in preparation for the flight experiments. These techniques include scanning electron microscopy and microprobe analysis of the distribution of silicon in silicon-germanium samples. Profilometry is being used to measure the surface roughness of the grown ingots. Ground based experiments will continue in 2015 in order to optimize the ampoule designs.
Bibliography Type:	Description: (Last Updated: 07/09/2021)
Abstracts for Journals and Proceedings	Volz MP, Mazuruk K, Croell A. "Influence of Containment on the Growth of Silicon-Germanium (ICESAGE): A Materials Science Investigation." Presented at the 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. Abstracts. http://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20140011618.pdf ; accessed 1/13/2015. , Jun-2014
Abstracts for Journals and Proceedings	Volz MP. "Materials Science in Microgravity." Presented at the 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. 3rd Annual International Space Station (ISS) Research and Development Conference, Chicago, Illinois, June 17-19, 2014. Abstracts. , Jun-2014

Articles in Peer-reviewed Journals	Volz MP, Mazuruk K. "Shape evolution of detached Bridgman crystals grown in microgravity." Crystal Research and Technology. 2014 Oct;49(10:768-76. http://dx.doi.org/10.1002/crat.201400116 , Oct-2014